Chan 2021 SCLC/NSCLC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → **GO**

24/34 samples (≥37 cells) share one continuous manifold, no per-patient islands (radius-graph: one component; iLISI median 1.00→8.8 of 24 samples at θ=10).

Gate 2 — module geography → **GO** (non-branching)

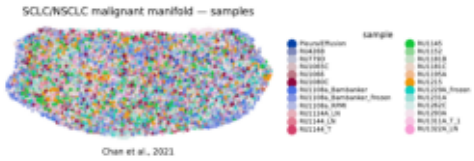
Single dominant pSom→Crest axis: pSom high over the neuroendocrine body, Crest anti-correlated (r=−0.89), rising to a terminal readout (crest-high decile mean pt=0.61). Gast and pSom co-vary (r=+0.28) — overlap, no competitive cross-repression.

Gate 3 — hub mutual exclusivity → **NA**

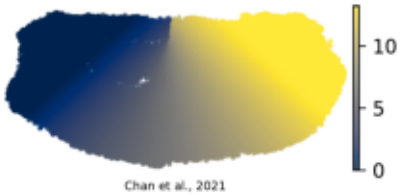
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

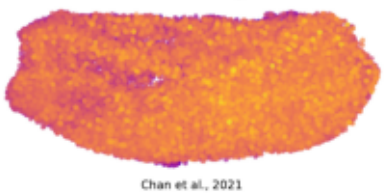
Gate 1: integration



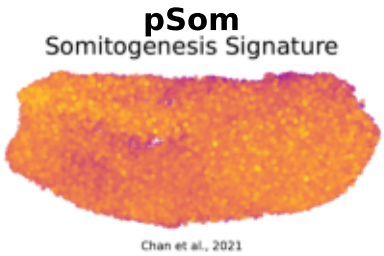
Trajectory / root



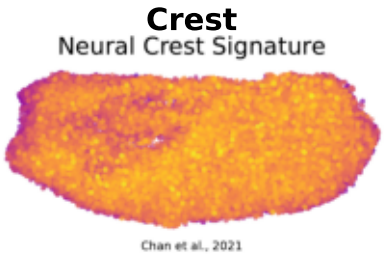
Gast



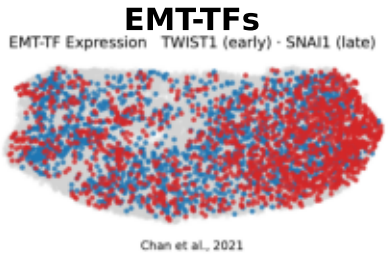
pSom



Crest



EMT-TFs



Dataset. Small-cell lung cancer (Chan et al., 2021), 10x — 34 samples / 51,052 malignant cells; predominantly SCLC (ASCL1/NEUROD1/POU2F3 neuroendocrine subtypes) with minor NSCLC. 24 samples (≥37 cells) → 50,997 cells enter the trajectory; the whole malignant manifold (no lobe scoping).

Verdict.
✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The SCLC malignant manifold is a single monotone somitogenesis→terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. A PPT tree over-forks the compact cloud (8–15 forks, no single dominant Y), while an elastic curve is fold-free (self-approach 2.78); Gast and pSom co-vary rather than competing. The developmental axis is orthogonal to proliferation (module–g2m |ρ|≤0.15), so the whole manifold is kept (no scoping). Caveat: pseudotime co-varies negatively with cycling (ρ≈−0.65), as the pSom-high root pole is the more proliferative one (Zhou/Bischoff precedent). EMT-TFs TWIST1 (early/root) → SNAI1 (late/terminus); ZEB1 is the most-expressed factor but pseudotime-flat; end-separation modest (Δpt=0.15).

Parameter choices

Min cells/sample	≥37 (24/34; gap 92→15)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=30 λ=0.3 μ=100
Root node	tip 2 = pSom/Gast-high, Crest-low	**EMT-TFs**	TWIST1 (early) → SNAI1 (late)
Geometry	Linear-pSom/Crest (non-branching)	**n_pcs**	20 (neighbours + iLISI)